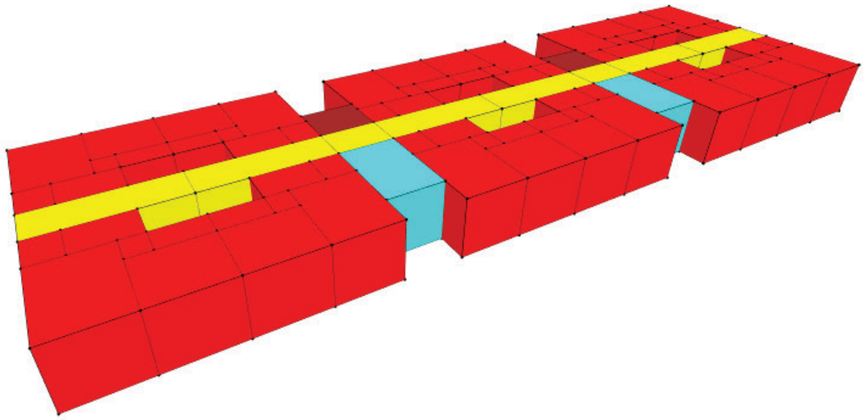
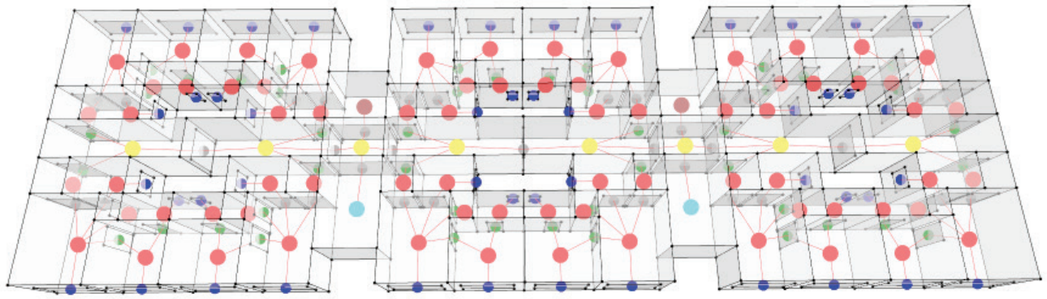
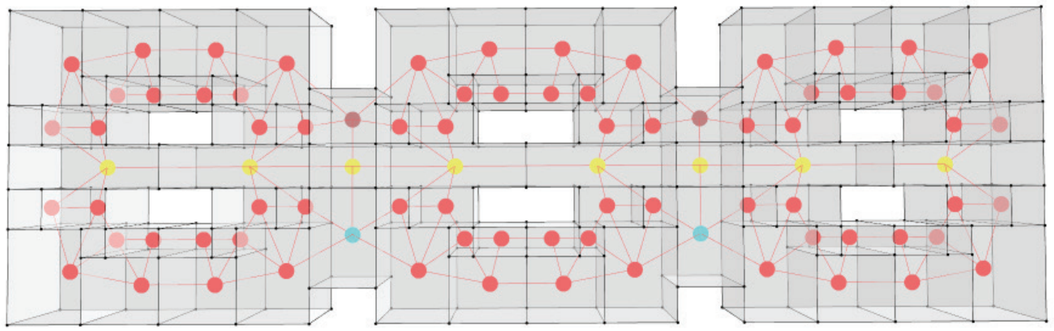


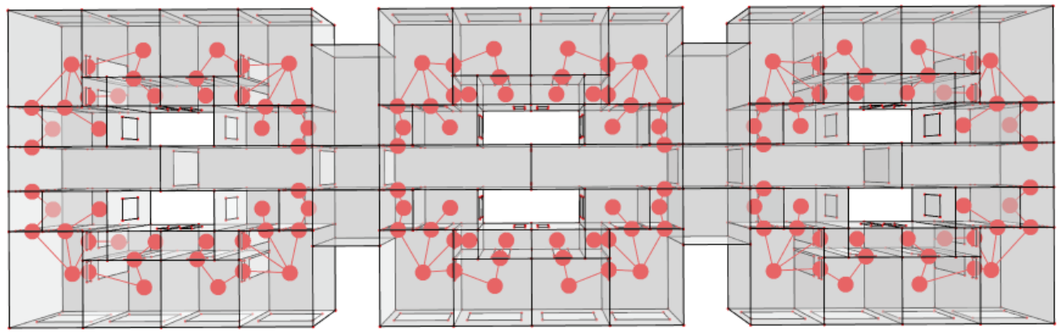
Building Massing: A color-coded 3D cell complex was constructed using CellComplex.ByCells(), organizing the apartment floor plan into discrete spatial units. Room dictionaries were assigned using Topology.TransferDictionariesBySelectors(), with Level 0 rooms rendered in red, corridors in yellow, stairways in cyan, and service spaces in brown.



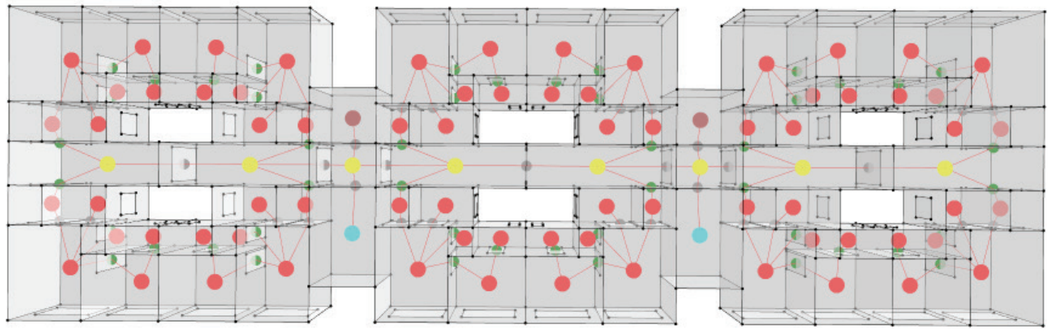
Graph Through Apertures: A connectivity graph was produced using Graph.ByTopology(apartment, viaSharedApertures=True, toExteriorApertures=True), incorporating doors, windows, and openings loaded from Doors.obj, Windows.obj, and Openings.obj as apertures. This graph captures both internal and external spatial connections through physical openings.



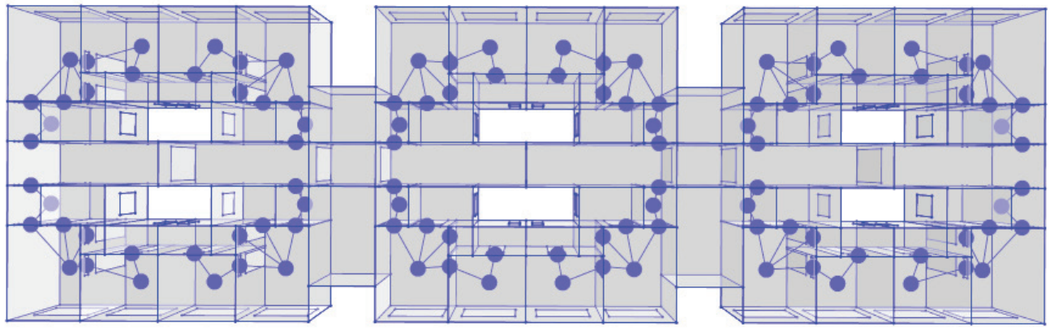
Adjacency Graph: A spatial adjacency graph was generated using Graph.ByTopology(apartment), where each node represents a room and edges indicate directly shared wall boundaries between neighboring cells. Node size and color are driven by room dictionary attributes.



Air Exhaust Diagram: An HVAC exhaust network was modeled by adding Ducts.obj as face apertures using Topology.AddApertures() and generating graph g_exh with viaSharedApertures=True. The resulting graph connects all rooms across the floor plan to the exhaust duct network, representing air extraction pathways.



Circulation Graph: An internal circulation graph was derived using toExteriorApertures=False, limiting connectivity to interior apertures only. This isolates the movement network within the building, excluding any connections to exterior openings.



Air Intake Diagram: An air intake network was derived from the same duct geometry, with Graph.RemoveVertex() applied to filter the graph to only retain Room_1, Room_2, and Room_5 as designated fresh air supply zones. This targeted graph represents the selective intake distribution across specific spaces in the floor plan.